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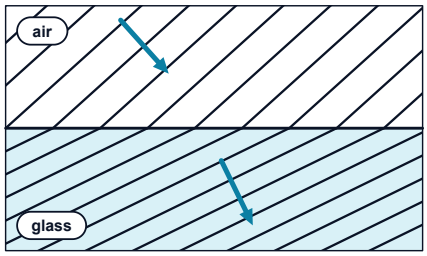
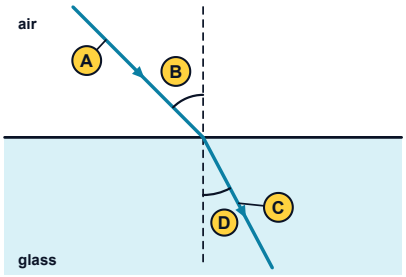
Electromagnetic waves

Waves
Paper 2 · Combined and Separate
32 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 16 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Light passes from air into glass. Compare the angle of refraction with the angle of incidence.**	It is smaller (the ray bends towards the normal).	☐
2	**Light passes from glass into air. Compare the angle of refraction with the angle of incidence.**	It is larger (the ray bends away from the normal).	☐
3 ★	**What type of wave are electromagnetic waves?**	Transverse	☐
4	**State what oscillates in electromagnetic waves.**	Electric and magnetic fields	☐
5	**Explain why electromagnetic waves can travel in a vacuum.**	They are oscillations of fields, not particles, so they need no medium.	☐
6 ★	**Name the seven groups of the electromagnetic spectrum, from longest to shortest wavelength.**	Radio waves, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays	☐
7	**Name the electromagnetic waves with the longest wavelength.**	Radio waves	☐
8	**Name the electromagnetic waves with the shortest wavelength.**	Gamma rays	☐
9	**Name the electromagnetic waves that human eyes can detect.**	Visible light	☐
10 ★	**Define refraction.**	The change in direction of a wave at a boundary between two media.	☐
11 ★	**Give two situations in which a wave crossing a boundary does not change direction.**	When it travels along the normal. When its speed is the same in both media.	☐

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	QUESTION	ANSWER	✓
12 ★	**Explain why refraction happens.**	The wave changes speed in the new medium.	☐
13	**Name the process shown in this wavefront diagram.** 	Refraction	☐
14	**State what labels A to D show in this diagram.** 	A: incident ray B: angle of incidence C: refracted ray D: angle of refraction	☐
15 ★	**Describe how radio waves can be produced.**	By oscillations in an electrical circuit.	☐
16 ★	**Describe how radio waves can be detected.**	They are absorbed by an aerial, creating an alternating current of the same frequency.	☐
17 ★	**State two hazards of ultraviolet radiation.**	Premature skin ageing and a higher risk of skin cancer	☐
18 ★	**State the hazards of X-rays and gamma rays.**	They are ionising: they can cause gene mutations and cancer.	☐
19	**State what radio waves are used for.**	Television and radio signals	☐
20 ★	**State what microwaves are used for.**	Satellite communications and cooking food	☐
21 ★	**State what infrared radiation is used for.**	Electrical heaters, cooking food and infrared cameras	☐
22	**State what visible light is used for.**	Fibre optic communications	☐

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	QUESTION	ANSWER	✓
23 ★	**State what ultraviolet radiation is used for.**	Energy-efficient lamps and sun tanning	☐
24 ★	**State what X-rays and gamma rays are used for.**	Medical imaging and treatments	☐
25 ★	**State what all electromagnetic waves have in common in a vacuum.**	They travel at the same speed.	☐
26	**Name the electromagnetic waves with the highest frequency.**	Gamma rays	☐
27	**State what electromagnetic waves transfer.**	Energy, from a source to an absorber.	☐
28	**State the kind of spectrum that electromagnetic waves form.**	A continuous spectrum	☐
29	**State where gamma rays originate.**	From changes in the nucleus of an atom.	☐
30 ★	**Explain why microwaves are suitable for satellite communications.**	They pass through the atmosphere without being absorbed or reflected.	☐
31 ★	**Explain why X-rays are suitable for producing images of bones.**	They pass through soft tissue but are absorbed by bone.	☐
32	**State two things that the hazard from UV, X-rays and gamma rays depends on.**	The type of radiation and the size of the dose	☐

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